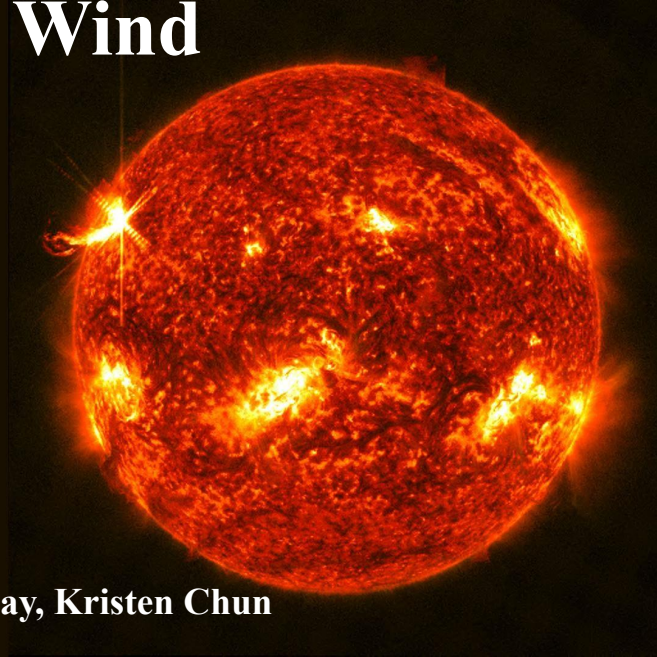


Heavy Ion Composition of Magnetic Switchback Patches in Solar Wind



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Mentor: Yeimy Rivera

Co-author: Samuel Badman

Science Research Mentoring Program Symposium

May 28th, 2025



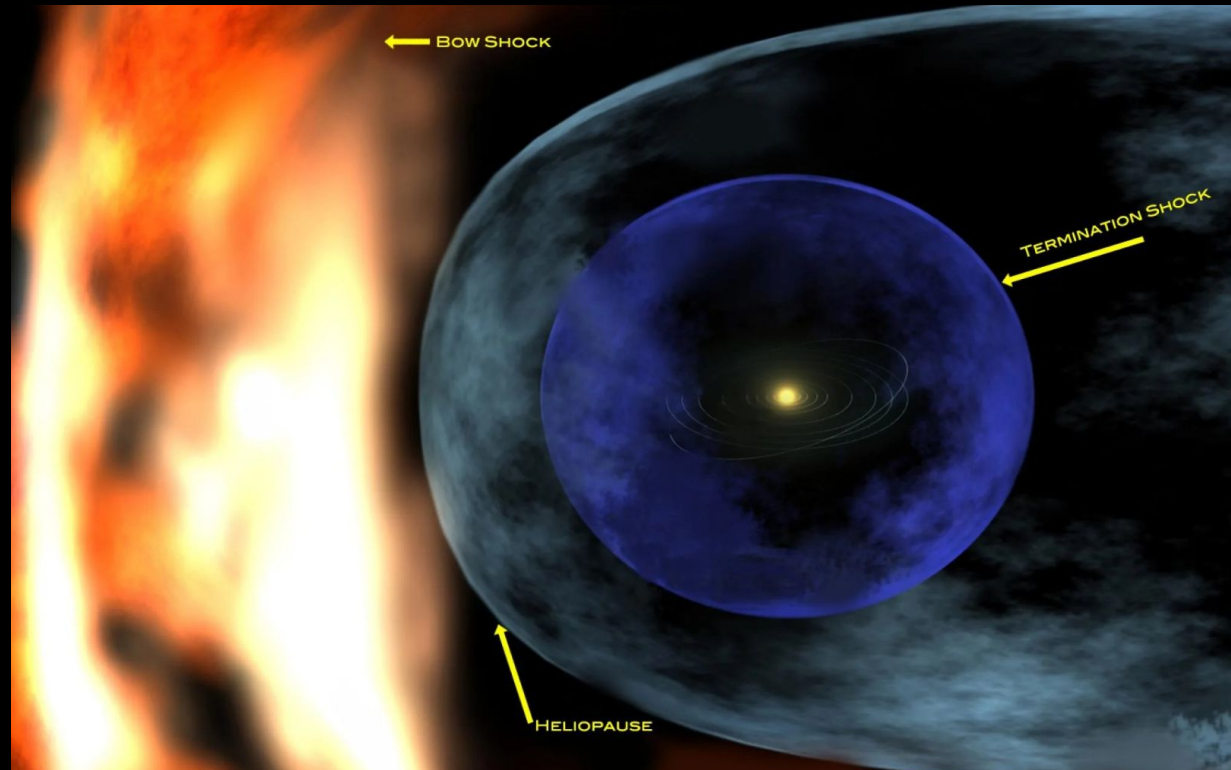
CENTER FOR

ASTROPHYSICS

HARVARD & SMITHSONIAN

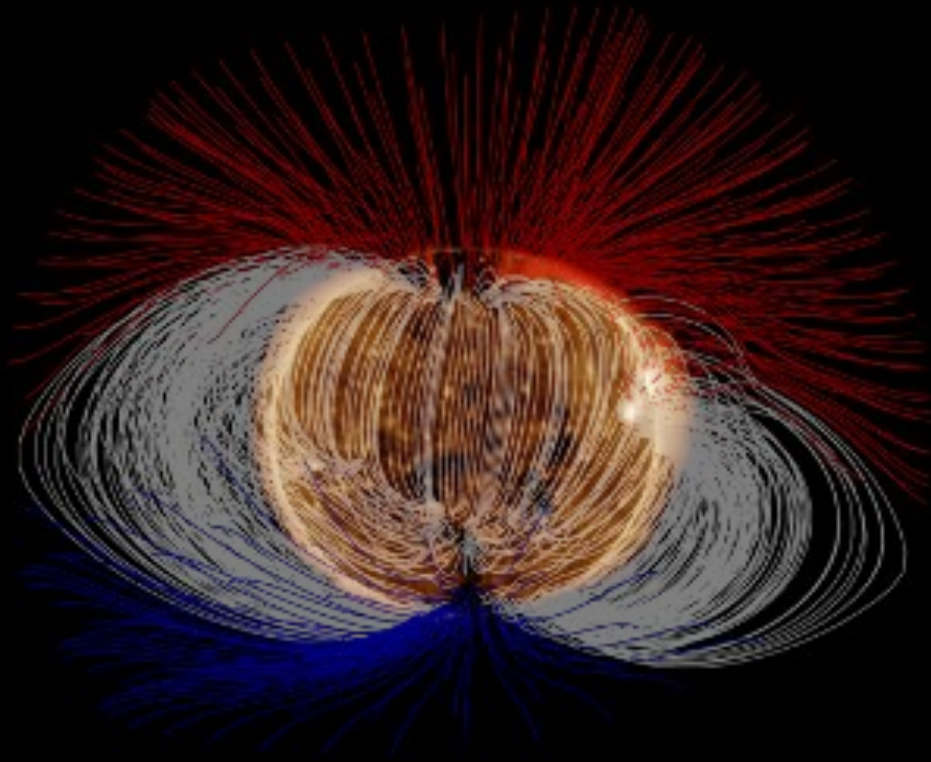
What is Solar Physics?

- Astrophysics subset → studies the Sun and heliosphere
- Types of solar wind: **fast (open field lines)** or **slow (closed field lines)**
- **Remote sensing** and **in situ measurements** allow us to study the solar wind

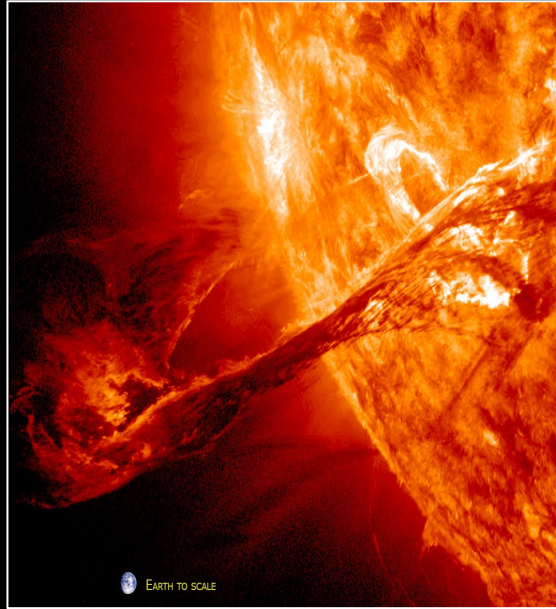


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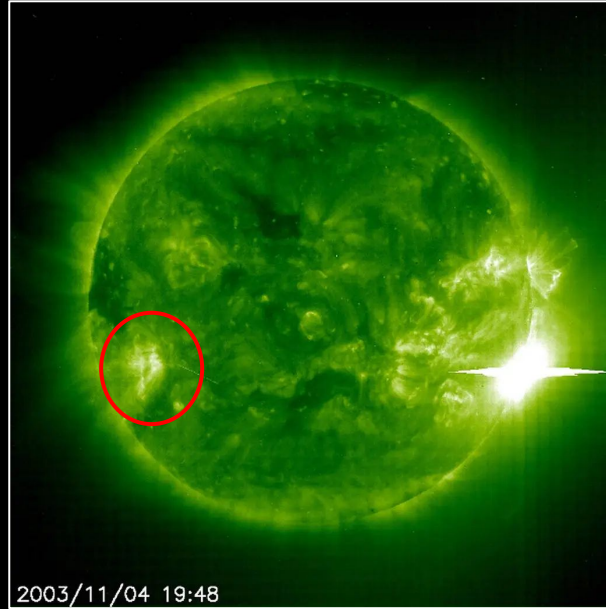
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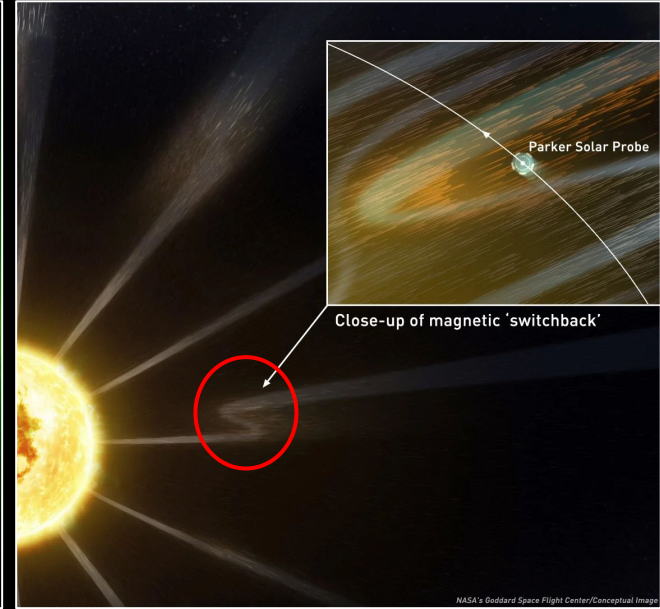
What do the Measurements Tell Us?



Coronal mass ejections



Solar flares



Magnetic switchbacks

How are Magnetic Switchbacks Formed?

National Aeronautics and Space Administration

Where Do Switchbacks Come From?

Switchbacks are sudden reversals in the solar wind's magnetic field. They were a surprise discovery as NASA's Parker Solar Probe made its first close flyby of the Sun in November 2018.

How do switchbacks form? Here are the current theories competing to explain them.

(Not to scale)
www.nasa.gov

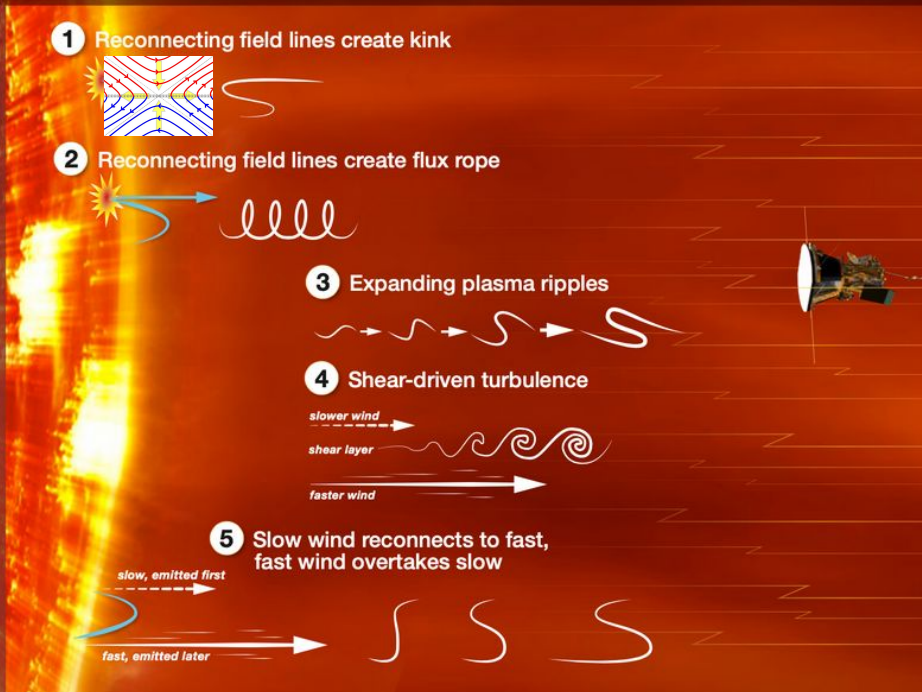
1 Reconnecting field lines create kink

2 Reconnecting field lines create flux rope

3 Expanding plasma ripples

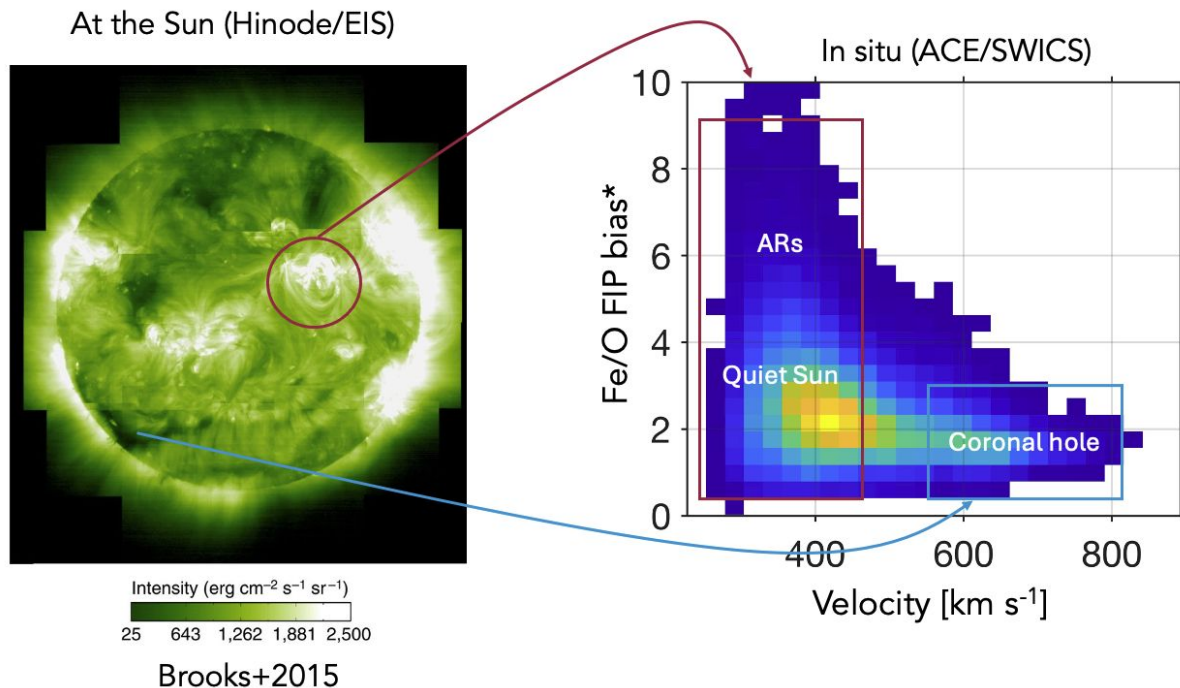
4 Shear-driven turbulence

5 Slow wind reconnects to fast, fast wind overtakes slow



We do not know how magnetic switchbacks are formed... BUT there are a few theories

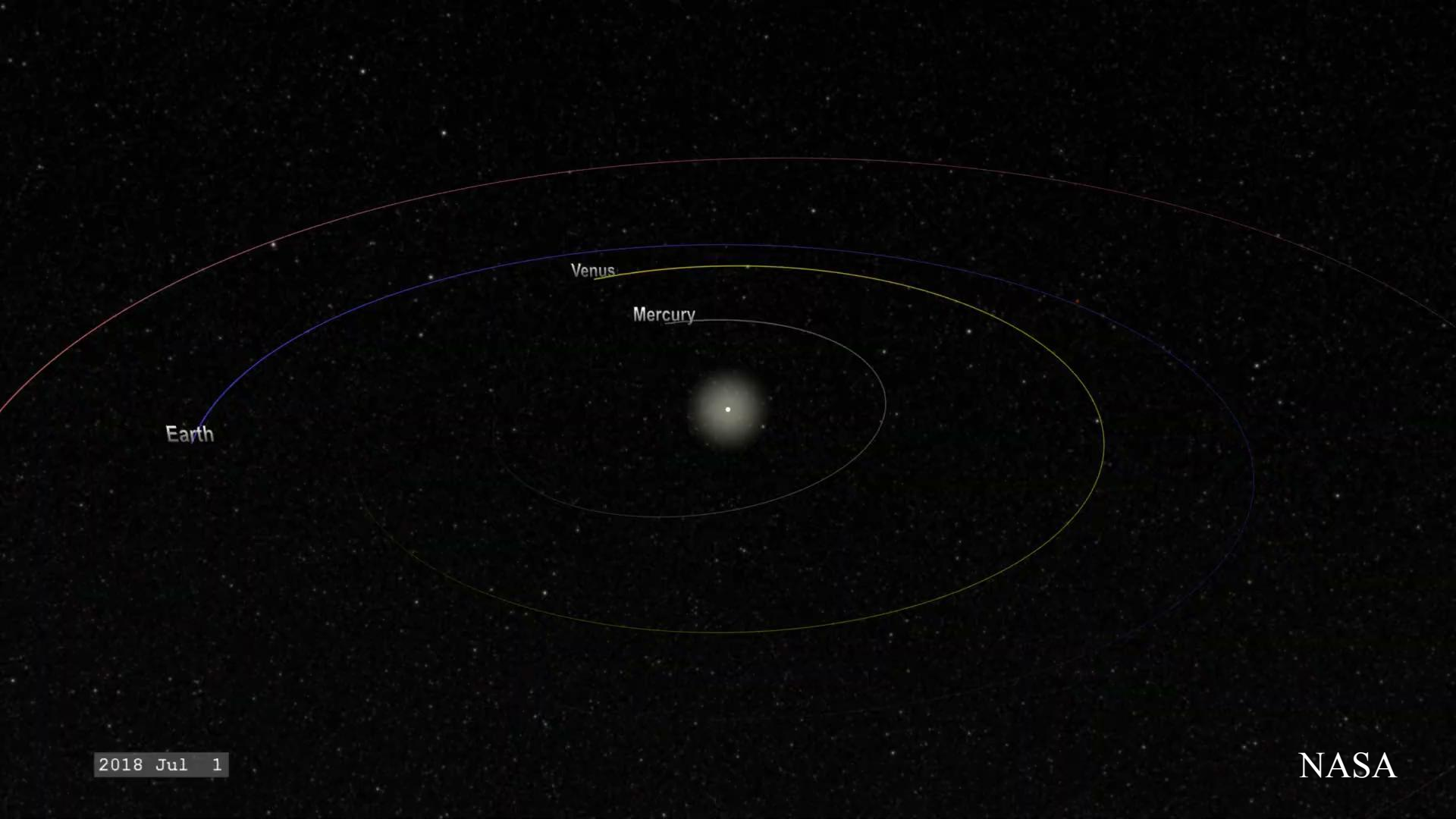
Focus of Our Work



- **Ion ratios** ($\text{C}^{6+}/\text{C}^{5+}$, $\text{O}^{7+}/\text{O}^{6+}$) gives insight into the temperature at the solar source
- **Elemental composition** (Fe/O) points to source region as open/closed magnetic topology

(Laming+2012, 2019)

Our goal: to determine where switchback patches form by examining **ion and elemental composition**



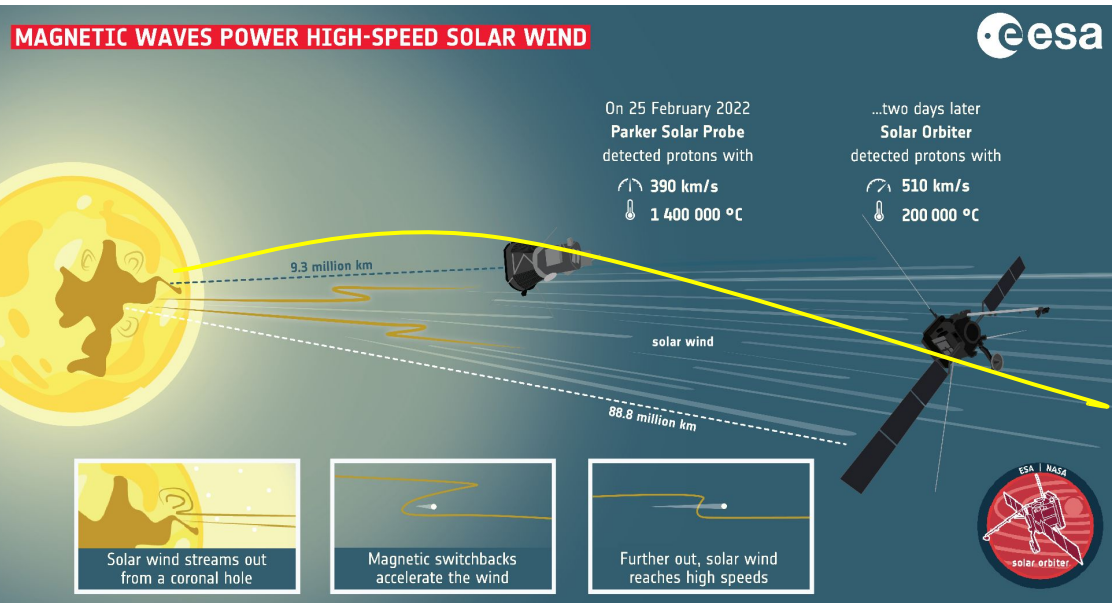
Venus

Mercury

Earth

2018 Jul 1

NASA



Why Parker Solar Probe and Solar Orbiter?

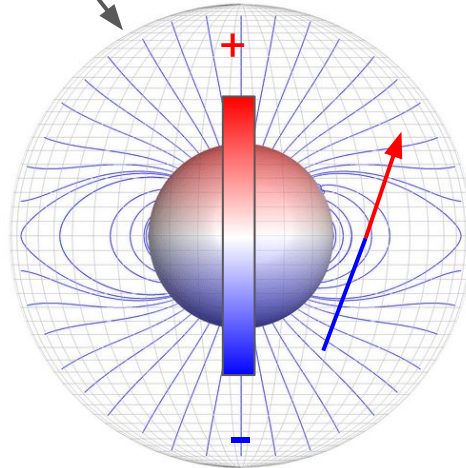
Limitations:

- **Parker** MUST be **at perihelion**
- **Solar Orbiter** has to **line up** with **Parker**
- Limitations to the compositional data: only spans January to April 2023 with **many gaps**
- No transients just **ambient solar wind**; NO coronal mass ejections
- **4 encounters—just two worked!**
 - We present 1 and Rivera+2024 examined 1 other

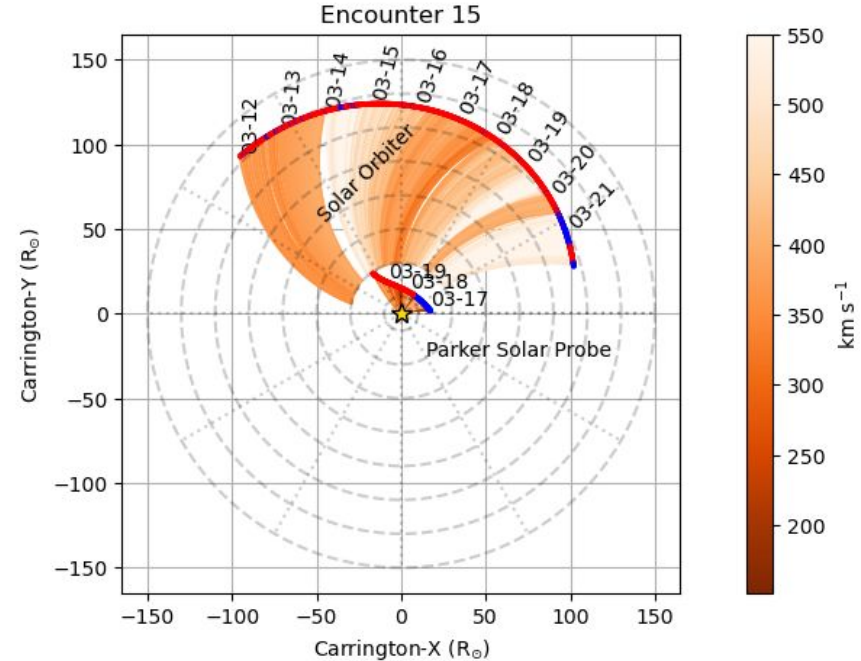
We need **Parker** to observe **magnetic switchbacks** and **Solar Orbiter** to collect **heavy ion data**.

Connecting Parker and Solar Orbiter

- We **backmap data points** from solar orbiter and parker solar probe to to **source surface** (Stansby+2019, Badman+2020)
- **At the source surface**, we can **compare the SAME data** from Parker AND Solar Orbiter.
- We **check our projections** with **current-sheet crossings**: polarity changes in the solar wind.

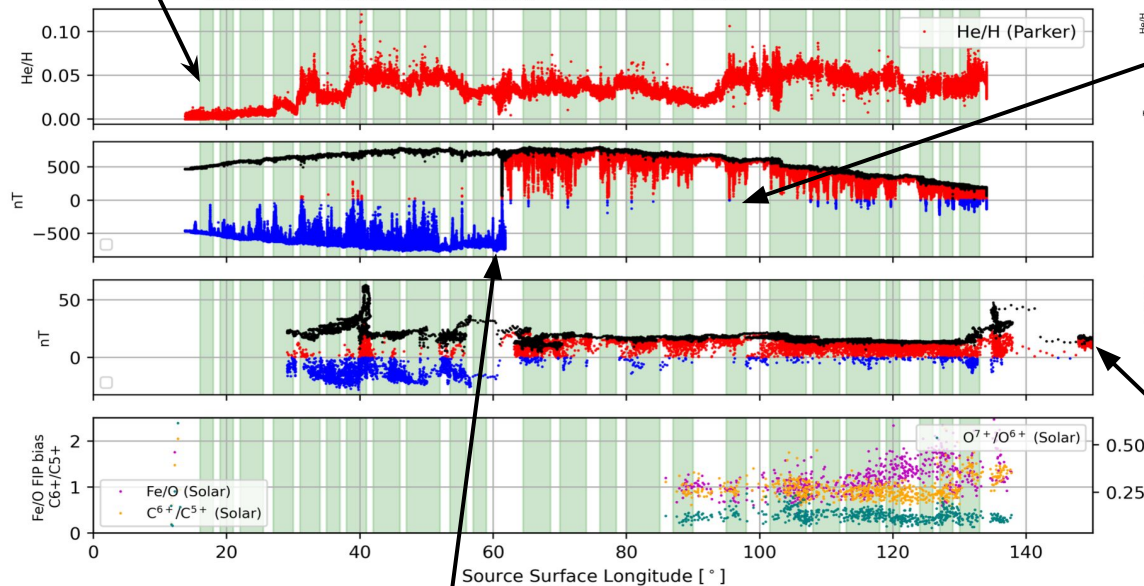


Birds-Eye View



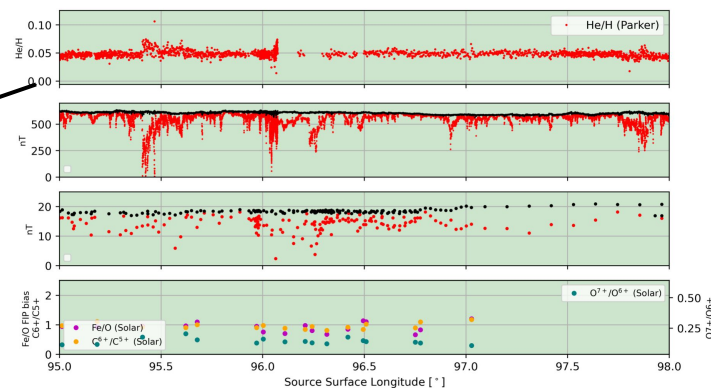
Switchback patch

Solar Orbiter & Parker Solar Probe Encounter 15



Current sheet crossing

Zoomed in Switchback Patch

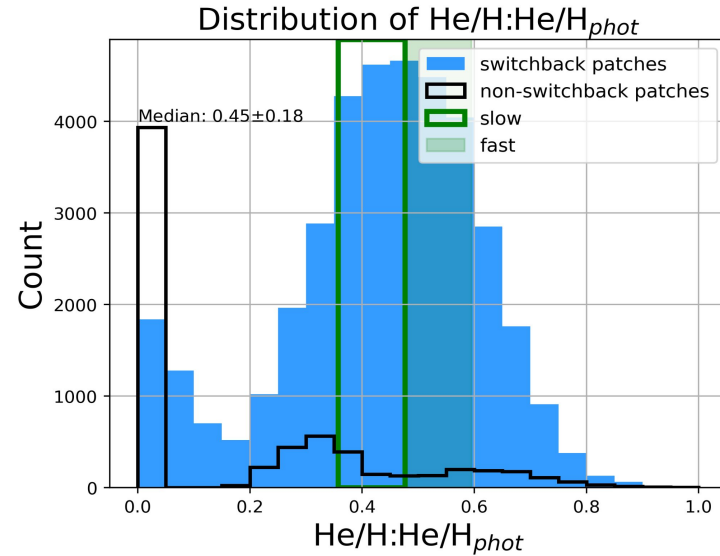
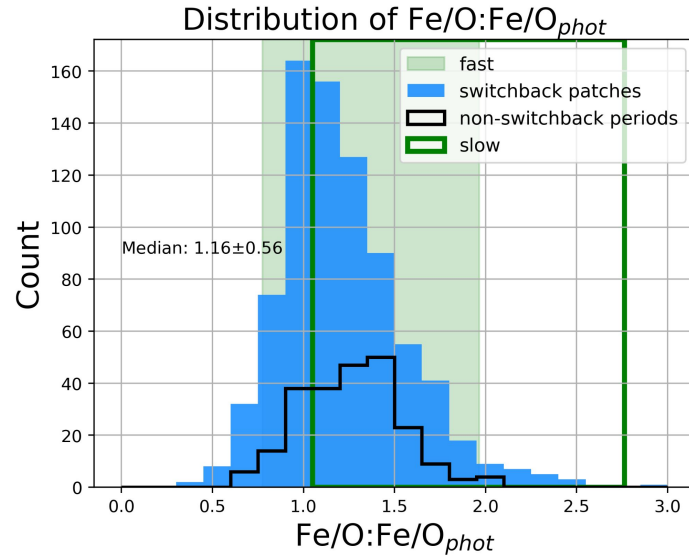


- = $+B_R$ (Parker & Solar Orbiter)
- = $-B_R$ (Parker & Solar Orbiter)
- = B Magnitude (Parker & Solar Orbiter)

Switchbacks (or kinks in the magnetic field)

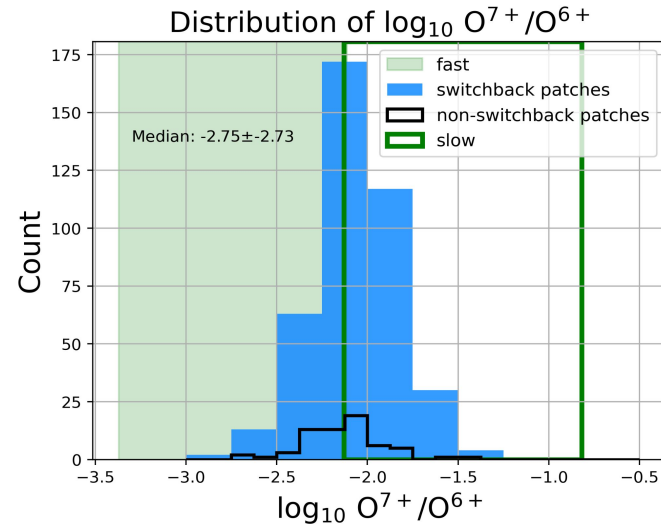
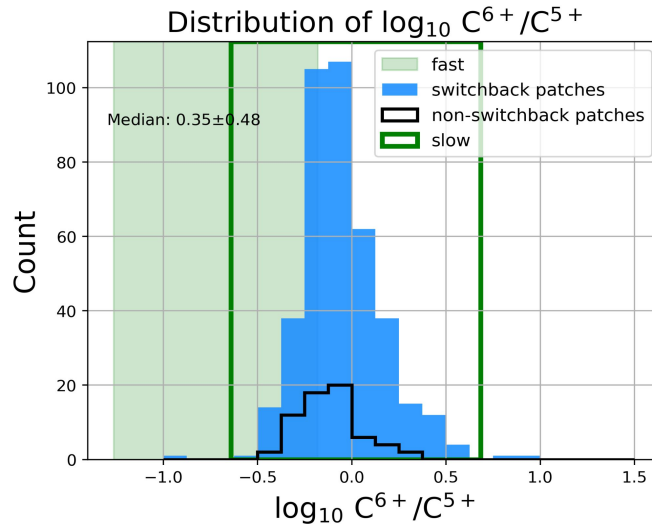
- Correlation between magnetic field and velocity
- Constant magnetic magnitude across the fluctuation
- Not a current sheet crossing

Elemental Composition of Switchbacks Patches



The elemental composition Fe and He suggest the switchback patches are coming from **open/closed fields**, with **fast and slow winds**

Ion Ratios of Switchbacks Patches



Ion ratios suggest a **hotter source than typical fast solar wind coming from open field**

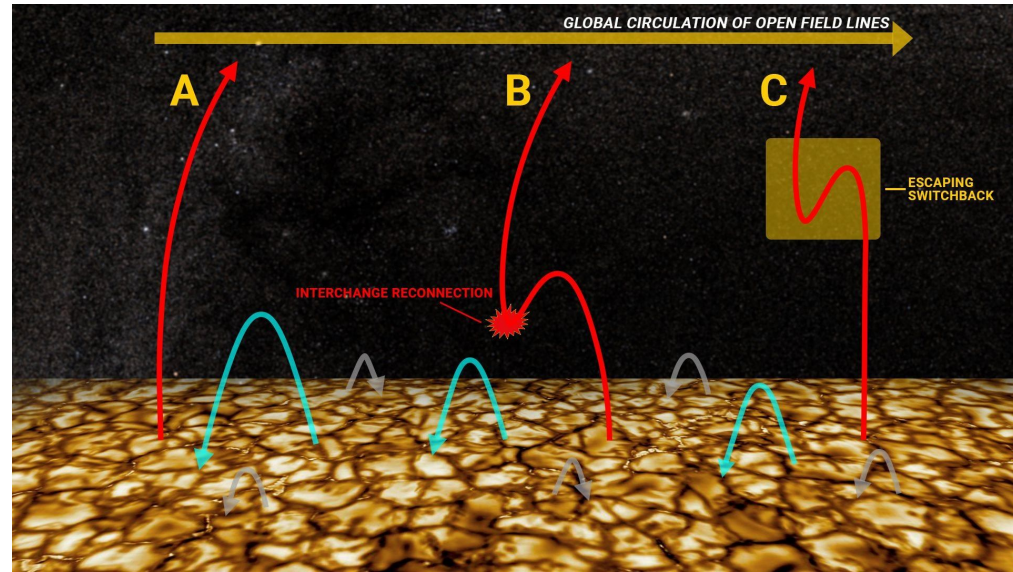
- Switchback patches do not reflect 'typical' coronal hole sources
- Additional heating is experienced at solar source as indicated by the solar source
- **Interchange reconnection** – drive some heating low in the corona (before solar wind leaves the sun)

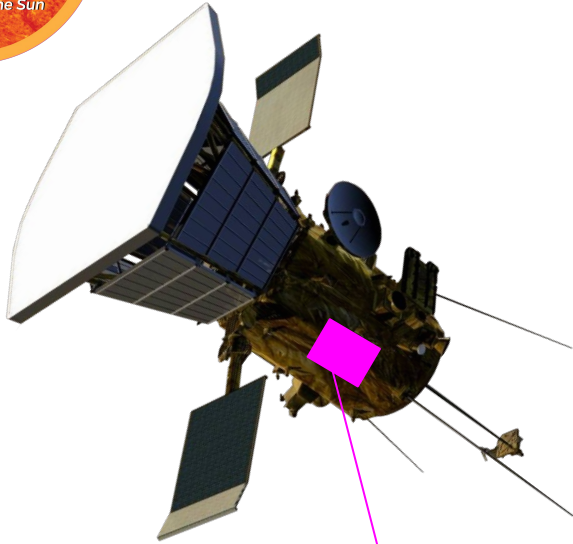
Main Takeaway

Summary: We **examined heavy ion properties** (Fe/O, C6+/C5+, O7+/O6+, He/H) of the solar wind around **Parker's 14th and 15th perihelion**, capitalizing on a spacecraft lineup with Solar Orbiter over a large range of longitude.

Results: Elemental composition (He and Fe) suggests fast wind composition.

Ion ratio suggests that it is **warmer than typical fast wind** flowing out of the coronal hole potentially due to **interchange reconnection (base heating)** at the source.





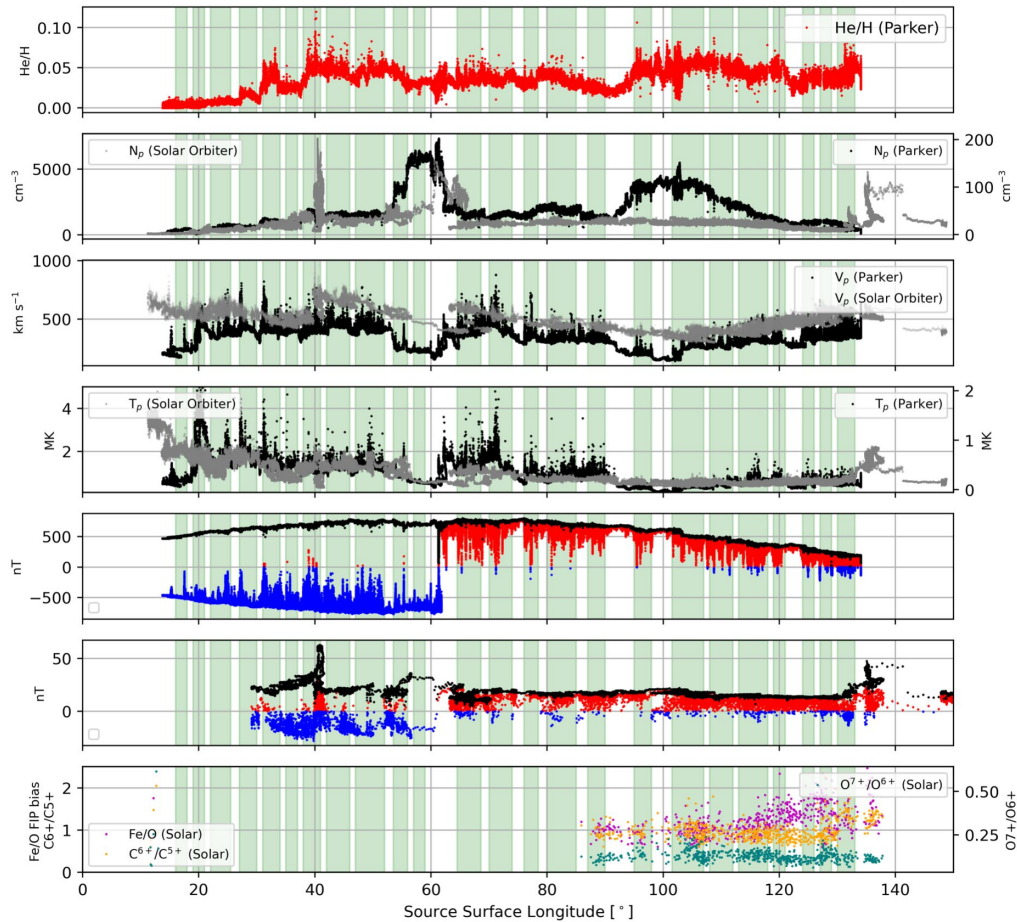
Heavy ion
compositional
instrument

Limitations:

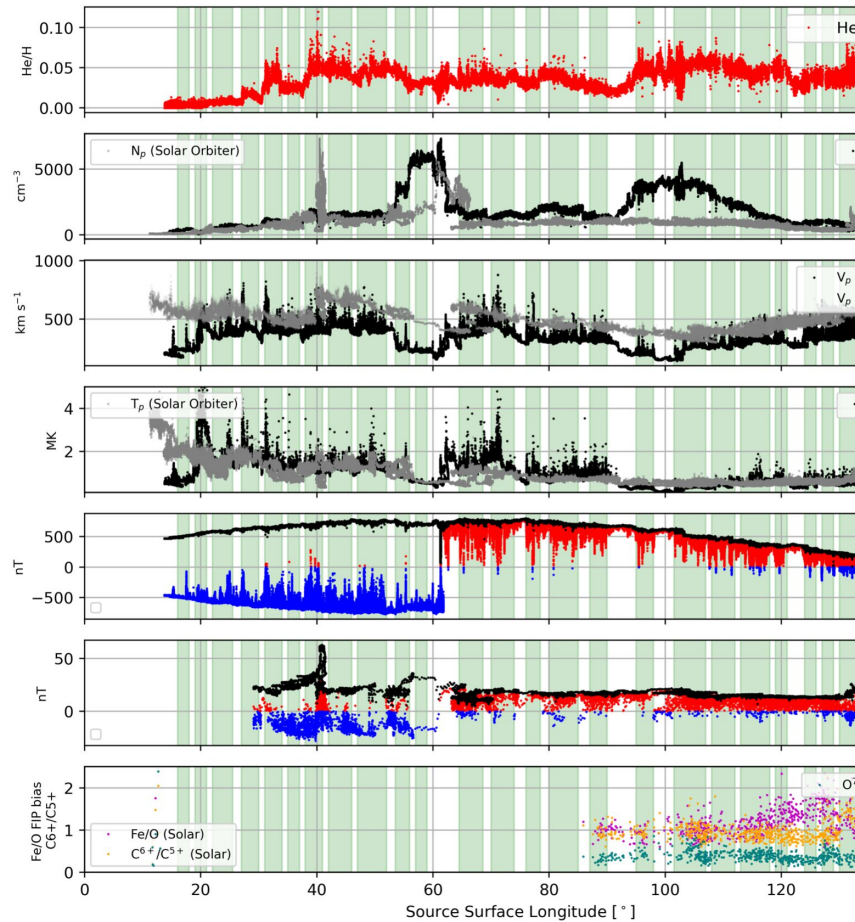
- **Two out of four** usable conjunctions.
- Fuel, cost, and weight are barriers.
- Not every instrument can look at what is flowing from the sun
- A special shield is needed so it does not melt during perihelion time
- Many types of reconnection (Flux ropes, interchange reconnection, jets/jetlets), but we would need a higher resolution to decipher
- The **helium abundance and ion composition does not tell the type of formation process.**

Future Outlook: We could potentially measure composition in the magnetic switchback itself right where it is at

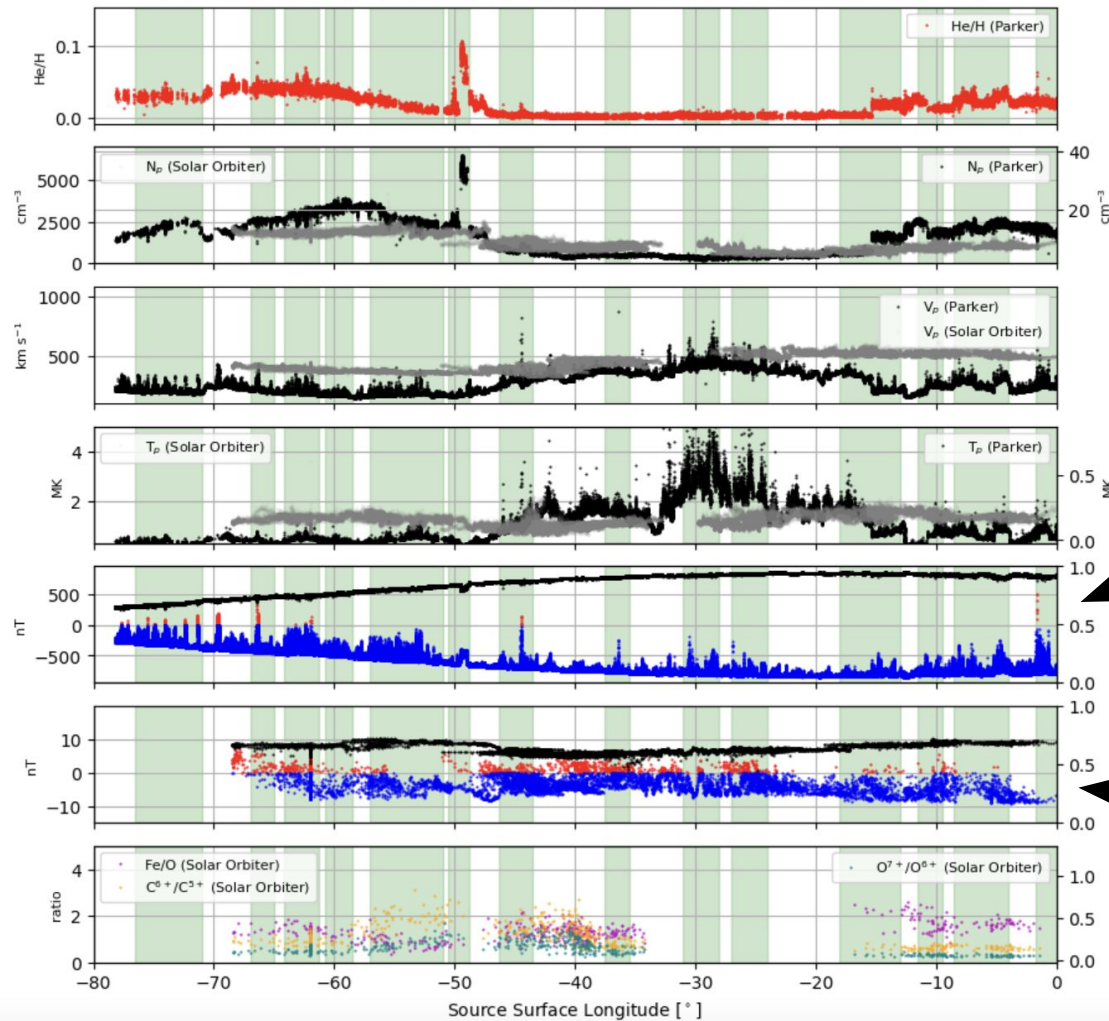
Solar Orbiter & Parker Solar Probe Encounter 15



Solar Orbiter & Parker Solar Probe Encounter 15



Parker Solar Probe & Solar Orbiter Encounter 14



Heliospheric properties of switchbacks

■ = $+B_R$ (Parker)

■ = $-B_R$ (Parker)

■ = B_{TOT} (Parker)

■ = B_R (Solar Orbiter)

■ = B_R (Solar Orbiter)

■ = B_{TOT} (Solar Orbiter)